

```

4.
5. def update_risk_parameters(market_volatility):
6.     # Tell Python we want to modify the global variables
7.     global position_size
8.     global stop_loss_percent
9.
10.    if market_volatility > 0.2: # High volatility
11.        position_size = 50 # Reduce position size
12.        stop_loss_percent = 0.03 # Tighter stop loss
13.    else:
14.        position_size = 100 # Standard position size
15.        stop_loss_percent = 0.05 # Standard stop loss
16.

```

While this capability exists, most professional trading systems avoid direct modification of globals. Instead, they use return values to provide updated information, which the caller can then decide how to use.

Functions in Action: Building Trading Components

Let's see how functions form the building blocks of real trading systems by creating a few common components used in algorithmic strategies.

Technical Indicator Functions

Technical indicators form the foundation of many trading strategies. Here's a function to calculate the Relative Strength Index (RSI), a popular momentum indicator:

```

1. def calculate_rsi(prices, period=14):
2.     # Calculate price changes
3.     changes = []
4.     for i in range(1, len(prices)):
5.         changes.append(prices[i] - prices[i-1])
6.
7.     # Separate gains and losses
8.     gains = [change if change > 0 else 0 for change in changes]
9.     losses = [-change if change < 0 else 0 for change in changes]

```